

Extended Mean-Field Theories for Networks of Real Neurons

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If the behavior of a system with many degrees of freedom can be captured by a small number of collective variables, then plausibly there is an underlying mean-field theory. We show that simple versions of this idea fail to describe the patterns of activity in networks of real neurons. An extended mean-field theory that matches the distribution of collective variables is at least consistent, though shows signs that these networks are poised near a critical point, in agreement with other observations. These results suggest a path to analysis of emerging data on ever larger numbers of neurons.

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Neurons in the brain generate discrete action potentials or spikes; in small windows of time, the responses are essentially binary, as each cell generates one spike or remains silent. The instantaneous state of a neural population thus is equivalent to the configuration of an Ising model, and we would like to know the distribution out of which these spin configurations are drawn. There is a long history of using ideas from statistical physics to approach this problem [1]; landmarks include the Hopfield model [2–4] and the Boltzmann machine [5], despite their limited biological realism. Both models are described by pairwise interactions among the Ising spins, and learning rules adjust these interactions to reach desired behaviors.

There is a direct path from experimental data to pairwise Ising models [1,6]. Concretely, we write the state of each neuron as s_n , with $s_n = +1$ when the cell is active and $s_n = -1$ when the cell is silent; the state of the network is $\mathbf{s} \equiv \{s_1, s_2, \dots, s_N\}$. We are looking for the probability distribution $P(\mathbf{s})$ and insist that it matches the observed means and correlations,

$$\langle s_n \rangle_P = \langle s_n \rangle_{\text{exp}}, \quad \langle s_n s_m \rangle_P = \langle s_n s_m \rangle_{\text{exp}}, \quad (1)$$

where $\langle \dots \rangle_P$ denotes an average over the distribution P and $\langle \dots \rangle_{\text{exp}}$ denotes an average over the data. Infinitely many distributions satisfy these constraints, but one is special because it generates states that are as random as possible and hence introduces no knowledge or structure beyond that required to match the expectation values in Eq. (1); this is the maximum entropy distribution [7,8]

$$P_{\text{pairs}}(\mathbf{s}) = \frac{1}{Z_{\text{pairs}}} \exp[-E_{\text{pairs}}(\mathbf{s})], \quad (2)$$

$$E_{\text{pairs}}(\mathbf{s}) = -\sum_n h_n s_n - \frac{1}{2} \sum_{n,m=1}^N s_n J_{nm} s_m. \quad (3)$$

We recognize this as the Boltzmann distribution for a particular Ising spin glass in which the fields $\{h_n\}$ and couplings $\{J_{nm}\}$ are determined by the data through Eq. (1); once we solve these equations all further predictions are parameter-free. These quantitative predictions have been very successful in several different systems [1].

The pairwise maximum entropy model takes $\sim N^2$ expectation values from the data. But current experimental methods allow dramatic increases in N without proportional increases in the duration T of experiments; eventually the measured matrix of pairwise correlations fails to be of full rank, limited by T and not N . As $N \rightarrow \infty$ at fixed T , we have enough data to measure $M \propto N$ expectation values, and then construct maximum entropy models consistent with these measurements. As an example, in an Ising model with local interactions we recover the correct Boltzmann distribution by asking for the maximum entropy model consistent with near-neighbor correlations, and there are $\mathcal{O}(N)$ of these even without translation invariance; reconstruction from local correlations also works in flocks of birds [9,10]. How do we choose $\mathcal{O}(N)$ constraints in the absence of guidance from symmetries or locality? The literature on activity in populations of neurons offers several intuitions.

Population activity—It has been suggested that collective effects in networks of neurons can be captured by keeping track of the summed activity [11–13],

$$\phi = \sum_{n=1}^N s_n. \quad (4)$$

The maximum entropy model that is consistent with the mean and variance of ϕ has the energy function

$$E_{\text{pop}}(s) = -h\phi - \frac{\lambda}{2N}\phi^2, \quad (5)$$

which we recognize as a mean-field ferromagnet [14–16]. Recall that in solving this model we rewrite it (exactly) as spins interacting with an auxiliary field. This connects to the intuition that the high dimensional dynamics of a network is controlled by a small number of “latent fields” to which the cells respond independently [17,18]; here the latent field is emergent rather than externally imposed. At large N fluctuations in the field are suppressed, and we refer to this as the naive mean-field approximation.

Means and projections—Focusing on the summed activity we lose the identities of the neurons. An alternative is to ask for a model that matches the mean activity of each individual cell, but tries to capture collective effects by looking only along one or a few projections

$$\varphi_\alpha = \frac{1}{\sqrt{N}} \sum_{n=1}^N W_{\alpha n} s_n \quad \alpha = 1, \dots, K, \quad (6)$$

where $\mathbf{W} \in \mathbb{R}^{K \times N}$. The maximum entropy model that matches all the mean activities $\langle s_n \rangle$ and the covariance matrix $\langle \varphi_\alpha \varphi_\beta \rangle$ again is a Boltzmann distribution [19–21]. This connects to the intuition that the high dimensional dynamics of a network are dominated by a low dimensional manifold [22–24]. Solving the model again introduces auxiliary fields, one conjugate to each projection φ_α , so this is a (slightly) generalized mean-field model [19]. For $K \ll N$ fluctuations in these fields are suppressed at large N , which again we refer to as naïve mean field. If $K \propto N$ there should be a more sophisticated mean-field theory analogous to the Hopfield network near saturation [25].

Distributions of projections—The emphasis on the variance of activity along particular projections may be too limiting. An alternative is to match the *distribution* of activity along a limited set of directions in the high dimensional space of neural activity; for simplicity we consider here just one projection φ . The maximum entropy model now is a Boltzmann distribution with energy

$$E_{\text{dist}}(s) = - \sum_{n=1}^N h_n s_n + NU(\varphi). \quad (7)$$

As usual the fields $\{h_n\}$ are adjusted to match the measured mean activity of each neuron and the potential $U(\varphi)$ is adjusted to match the observed distribution $P_{\text{exp}}(\varphi)$; the factor N keeps the potential of order one. This is equivalent to having a single latent field act on all the neurons, with an arbitrary distribution. It also connects to models for dense associative memory, or “modern Hopfield” networks [26].

Our major result is the construction of a generalized mean-field theory for this class of models and the demonstration that real networks are in the regime where this approximation is valid. In contrast, we will see that simpler mean-field theories fail.

Calculating the partition function of the population activity model [Eq. (5)] is a standard exercise [14–16]. The partition function can be rewritten exactly as

$$Z_{\text{pop}}(h, \lambda) = \sqrt{\frac{N}{2\pi\lambda}} 2^N \int d\psi e^{-Nf(\psi)}, \quad (8)$$

$$f(\psi) = \frac{1}{2\lambda} \psi^2 - \ln \cosh(h + \psi). \quad (9)$$

At large N the integral in Eq. (8) should be dominated by $\psi = \psi_*$, which minimizes the local free energy $f(\psi)$, and this leads to the naïve mean-field approximation

$$\ln Z(h, \lambda) = -Nf(\psi_*) + N \ln 2 - \frac{1}{2} \ln[2\pi\lambda f''(\psi_*)] + \dots, \quad (10)$$

where $\dots$ are terms $\sim 1/N$. In this approximation

$$\langle \phi \rangle \equiv \mu = \tanh(h + \lambda\mu) + \mathcal{O}(N^{-1}), \quad (11)$$

$$\frac{1}{N} \langle (\delta\phi)^2 \rangle \equiv \chi = \frac{1 - \mu^2}{1 - \lambda(1 - \mu^2)} + \mathcal{O}(N^{-1}). \quad (12)$$

These equations imply a maximum variance χ at fixed mean μ ,

$$\chi_{\text{max}}(\mu) = \frac{\mu(1 - \mu^2)}{\mu - \text{atanh}(\mu)(1 - \mu^2)}. \quad (13)$$

In Fig. 1 we show χ vs μ from data on several systems, and we see that the bound from Eq. (13) is violated, consistently. Robustness of these results to potential nonstationarities is explicitly assessed in the End Matter.

The conclusion from Fig. 1 is that real neurons are not consistent with naïve mean-field theory. But what happens to the maximum entropy problem? We still find values of h and λ in Eq. (5) such that the predicted μ and χ agree with measured values, but this must be done very carefully [20,21]. We can use the exact integral representation of Z_{pop} in Eq. (8) to construct, numerically, a trajectory $h_*(\lambda)$ along which $\mu = \mu_{\text{exp}}$ (Fig. 2, inset). This trajectory begins at $h_*(\lambda = 0) = \text{atanh}(\mu_{\text{exp}})$ and h_* moves toward $h_* = 0$ as λ increases. The trajectory stalls, with $dh_*/d\lambda$ becoming almost zero at a critical λ , at which point the susceptibility χ rises rapidly; in the examples from Fig. 1, χ_{exp} intersects this steep rise, so that λ is determined very precisely by the data.

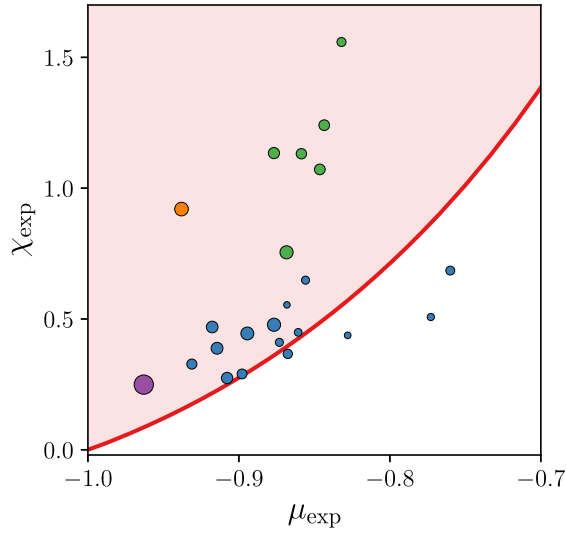


FIG. 1. Susceptibility and magnetization in real networks (points) vs bound from naïve mean-field theory (red line). Data from $N = 160$ neurons in the retina (orange), recorded with an electrode array [27]; $N = 1416$ neurons in the mouse hippocampus (purple), recorded with a fluorescent calcium indicator [28,29]; $N = 60$ – 190 neurons from single brain regions in a mouse (blue) and $N = 900$ – 1500 neurons across multiple regions (green), recorded with Neuropixels 2.0 [30].

Having found h and λ to match the mean and variance of the population activity ϕ , we can look directly at the local free energy $f(\psi)$ from Eq. (9); results are shown in Fig. 2 for a population of neurons in the mouse hippocampus [28,29]. There are two local minima, very nearly degenerate. The population of hippocampal neurons analyzed here lives in a plane, and so we can change N by expanding the radius of a circle [31,32]. As N increases the two minima of the local free energy become more nearly degenerate, and the local free energy difference scales as $1/N$ [20,21]. This is a defining feature of a first order phase transition. We emphasize that this is driven entirely by the experimental observations μ_{exp} and χ_{exp} .

The fact that the data drive these minimally structured models toward a first order transition is provocative. This result explains the failure of the naïve mean-field approximation and is suggestive of a network balanced between macroscopically ordered patterns of activity. Unfortunately, the presence of two local minima in $f(\psi)$ predicts that the distribution of summed activity will be bimodal, and this is qualitatively wrong.

These problems in matching the mean and variance of population activity persist in models that match the mean activity of individual neurons and variances of arbitrary projections, though the details depend on our choice of the projections. If we choose a single projection with coefficients W_n given by independent Gaussian random numbers, then the naïve mean-field approximation works but the resulting model has an entropy only slightly below that

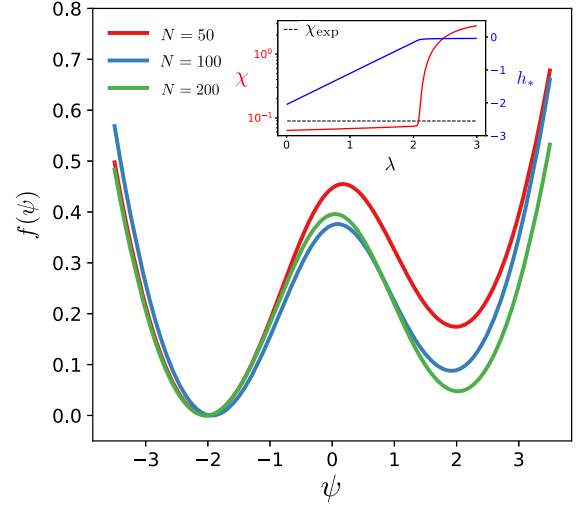


FIG. 2. Free energy per neuron $f(\psi)$ in the model that matches the mean and variance of summed population activity, Eq. (9), with h and λ inferred from experiments in the mouse hippocampus [28,29]. As N increases, the two local minima become more nearly degenerate, signaling the proximity of a first order phase transition. The inset shows the trajectory $h_*(\lambda)$ (blue) with constant magnetization and the corresponding susceptibility χ (red) for a network of $N = 50$ neurons; the intersection of χ with χ_{exp} (dashed line) determines the exact solution of the maximum entropy problem.

for completely independent neurons, with vanishing gap as $N \rightarrow \infty$ [20,21].

To order possible choices of the projection coefficients it is useful to think about the correlation matrix

$$\tilde{C}_{nm} \equiv \frac{\langle s_n s_m \rangle - \langle s_n \rangle \langle s_m \rangle}{[(1 - \langle s_n \rangle^2)(1 - \langle s_m \rangle^2)]^{1/2}}. \quad (14)$$

This matrix has eigenvectors $\tilde{W}$ and eigenvalues ρ . If we choose projections with coefficients

$$W_{an} = \frac{1}{\sqrt{1 - \langle s_n \rangle^2}} \tilde{W}_{an}, \quad (15)$$

then following the same naïve mean-field approximation that leads to Eq. (10) gives the entropy [19–21],

$$S = S_0 - \frac{1}{2} \sum_{\alpha} [\rho_{\alpha} - 1 - \ln(\rho_{\alpha})], \quad (16)$$

where S_0 is the entropy of independent neurons. Thus the most informative projections are those connected to the largest or smallest eigenvalues of the correlation matrix. But if we choose these optimized projections we are led back to the problem of degenerate local minima [20,21].

To tame the multiple minima we insist on matching both the mean activities of individual neurons and the *distribution* of activities along a projection, leading to Eq. (7).

The partition function (see End Matter),

$$Z_{\text{dist}} = 2^N \int \frac{dz}{2\pi} \int d\varphi \exp[-Nf_{\text{dist}}(\varphi; z)], \quad (17)$$

where the free energy

$$f_{\text{dist}}(\varphi; z) = U(\varphi) + \frac{1}{N} \left[iz\varphi - \sum_{n=1}^N \ln \cosh \left(h_n + iz \frac{W_n}{\sqrt{N}} \right) \right]. \quad (18)$$

As usual we search for a saddle point $(\varphi_{\text{sp}}, z_{\text{sp}})$ that extremizes $f_{\text{dist}}(\varphi; z)$; we find

$$\varphi_{\text{sp}} = \frac{1}{\sqrt{N}} \sum_{n=1}^N W_n \tanh \left(h_n + iz_{\text{sp}} \frac{W_n}{\sqrt{N}} \right), \quad (19)$$

$$z_{\text{sp}} = iNU'(\varphi_{\text{sp}}), \quad (20)$$

and approximate $\ln Z_{\text{dist}} \approx -Nf_{\text{dist}}(\varphi_{\text{sp}}; z_{\text{sp}})$. Notice that $E_{\text{dist}}(s)$ is invariant under the gauge transformation $U(\varphi) \rightarrow U(\varphi) - \varphi U'(\varphi_{\text{sp}})$ and $h_n \rightarrow h_n + \sqrt{N}U'(\varphi_{\text{sp}})W_n$, so we can set $U'(\varphi_{\text{sp}}) = 0$. This fixes $z_{\text{sp}} = 0$ and yields

$$\langle s_n \rangle = \frac{\partial Z_{\text{dist}}}{\partial h_n} = \tanh h_n, \quad (21)$$

as if the neurons responded independently to the fields

$$h_n = \text{atanh}(\langle s_n \rangle_{\text{exp}}). \quad (22)$$

The distribution of activity along the projection is

$$P(\varphi) = \frac{2^N}{Z_{\text{dist}}} \int \frac{dz}{2\pi} \exp[-Nf_{\text{dist}}(\varphi; z)]. \quad (23)$$

The saddle point approximation is

$$P(\varphi) \propto \exp[-Nf_{\text{dist}}(\varphi; z_{\star}(\varphi))], \quad (24)$$

where $z_{\star}$ depends on φ and is the solution of

$$\sum_{n=1}^N \frac{W_n}{\sqrt{N}} \tanh \left[h_n + i \frac{W_n}{\sqrt{N}} z_{\star}(\varphi) \right] = \varphi. \quad (25)$$

Given weights $\{W_n\}$ and the fields from Eq. (22), we solve Eq. (25) numerically to give $z_{\star}(\varphi)$. From Eq. (24), the free energy $f_{\text{dist}}[\varphi; z_{\star}(\varphi)]$ can be estimated directly from the measured activity along the projection; combining with Eq. (18) and solving for $U(\varphi)$ leads to

$$NU(\varphi) = \ln P_{\text{exp}} - i\varphi z_{\star}(\varphi) + \sum_{n=1}^N \ln \cosh \left(h_n + iz_{\star}(\varphi) \frac{W_n}{\sqrt{N}} \right). \quad (26)$$

We have applied this analysis to experiments on the mouse hippocampus, choosing the projection corresponding to the largest eigenvalue of the correlation matrix (Fig. 3). We see that the potential $U(\varphi)$ departs significantly from quadratic at large $|\varphi|$, hinting as to how this model can “work” even though fixing the variance of activity along the projection fails, as above.

To test our generalized mean-field theory we performed a Monte Carlo simulation with energy from Eq. (7), the potential from Fig. 3(a), and fields from Eq. (22). Figure 3(b) shows that we correctly reproduce the distribution of activity along the projection; the inset shows that we recover the correct mean activities as well. The troublesome local minima have been banished.

For some parameter values the curvature of the free energy vanishes at the saddle point, signaling a second order phase transition. The condition for this to happen is not simply $U''(\varphi_{\text{sp}}) = 0$, since the free energy also includes entropic contributions (see End Matter). This condition instead is $1 + NU''(\varphi_{\text{sp}})\Delta = 0$, where Δ is the variance of activity along the projection that we would find if the neurons were independent,

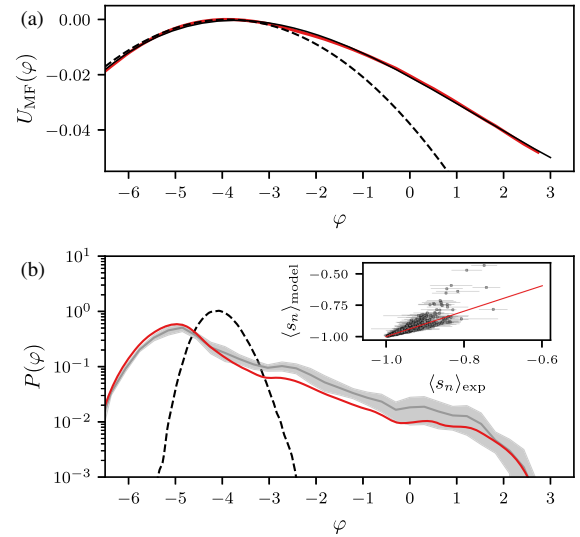


FIG. 3. Distributional maximum entropy model from Eq. (7), for $N = 1416$ neurons in the mouse hippocampus. The projection corresponds to the maximum eigenvalue of the correlation matrix. (a) The potential $U(\varphi)$ (solid red) compared with a quadratic (black dashed) and a cubic (black solid). (b) The distribution of activity $P(\varphi)$ from the model (red), the data (gray), and independent neurons (dashed). The experimental distribution is approximated by a histogram with $N_b = 32$ bins; shading shows s.d. across tenths of the data. Inset: mean activity of each neuron, model vs data.

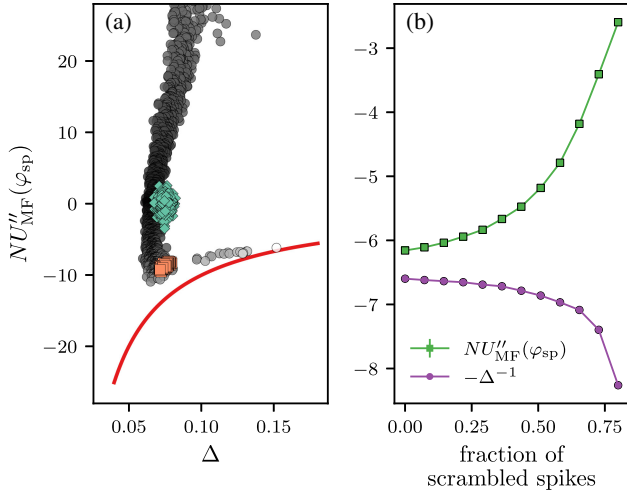


FIG. 4. Approach to criticality in the model of Fig 3. (a) Scatter plot of $NU''(\varphi_{\text{sp}})$ vs Δ for different projections: random (green), random positive (orange), and the eigenvectors (gray) of the correlation matrix as in Eq. (15), lighter colors correspond to large ΔS . The critical line $1 + NU''(\varphi_{\text{sp}})\Delta = 0$ is in red. (b) Trajectories of $NU''(\varphi_{\text{sp}})$ and $-\Delta^{-1}$ as we look at networks with same mean activity for each neuron but weaker correlations, generated by shuffling a fraction of the time bins independently for each neuron.

$$\Delta = \frac{1}{N} \sum_{n=1}^N W_n^2 [1 - \langle s_n \rangle^2]. \quad (27)$$

With random projections $NU''(\varphi_{\text{sp}})$ is near zero [20,21], far from criticality [Fig. 4(a)]; with random positive coefficients $NU''(\varphi_{\text{sp}})$ is consistently negative but still not close to $-\Delta^{-1}$. With projections along the eigenvectors of the correlation matrix [Eq. (15)], we see modes that trail toward the critical line, with the highest variance principal component lying within $\sim 8\%$ of criticality.

The approach to criticality depends on the strength of correlations. We can imagine systems in which the mean activity of each neuron is the same but correlations between pairs of neurons are weaker, and we can generate such data by shuffling a fraction of the time bins independently for each neuron. For each shuffled dataset we repeat the construction above, and we find that $NU''(\varphi_{\text{sp}})$ and $-\Delta^{-1}$ gradually move apart [Fig 4(b)]: plausible populations of neurons are farther from criticality than the real network. This is consistent with other signatures of near-critical behavior [1], including the scaling of these same data under coarse graining [29].

We can compute the entropy of the model in Eq. (7) in the mean-field approximation (see End Matter),

$$S = S_0 + N[\langle U(\varphi) \rangle - U(\varphi_{\text{sp}})] - \frac{1}{2} \ln [1 + NU''(\varphi_{\text{sp}})\Delta], \quad (28)$$

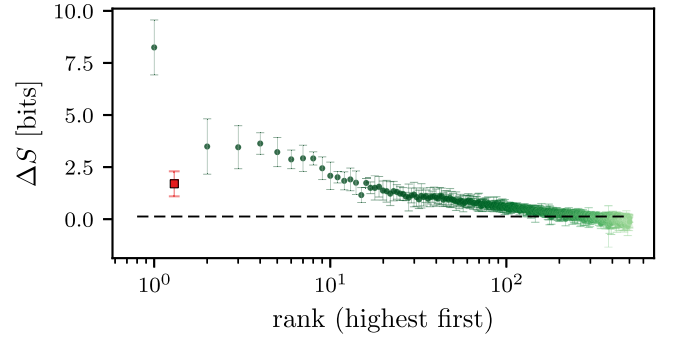


FIG. 5. Entropy reduction relative to independent neurons for the model in Eq. (7), using each of the 500 highest variance projections (green, larger ρ_α are darker) or projection onto summed activity (red square). The dashed line is the independent entropy per neuron $S_0/N \sim 0.13$ bits.

with results in Fig. 5 for the same data as in Fig. 4. Matching the distribution along just one projection leads to an entropy reduction of $\sim 5\%$ of the independent entropy. There are roughly 200 projections that each provide a contribution above the independent entropy per neuron. Entropy alone is not sufficient to assess the quality of a model; a good model should capture statistical structure beyond the observables it explicitly constrains [1]. Also in this regard, the model is successful (see End Matter).

To summarize, it proves surprisingly difficult to construct a consistent mean-field theory for patterns of activity in a network of real neurons. We finally succeed with a model that matches the mean activity of individual neurons and the *distribution* of activity along one projection. But there are many projections along which constraining the distribution generates a substantial reduction in entropy. Our approach is agnostic to the dynamics, capturing only the stationary distribution of neural activity. An important future direction is to extend this framework to models that constrain entire trajectories, which would give access not only to static but also to dynamical signatures of criticality [33], such as the divergence of the correlation time. An important future direction is thus to extend this approach to match the distribution of multiple projections, giving a framework for describing networks of N neurons with $\mathcal{O}(N)$ constraints and making it possible to analyze data at much larger N .

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Data availability—Some of the data that support the findings of this article are openly available [34]. Calcium imaging recordings of the CA1 mouse brain region, supporting Figs. 1–4 are not publicly available because they are owned by a third party and the terms of use prevent public distribution. The data are available from the authors upon reasonable request.

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End Matter

Stationarity concerns—Throughout this Letter we have assumed that the recorded neural activity is stationary and ergodic over the duration of each dataset. To empirically assess the validity of these assumptions, we divided each recording into two consecutive, equally sized temporal segments. For each segment we independently computed the experimentally measured observables χ_{exp} and μ_{exp} . As shown in Fig. 6, we find a strong correlation between the values of χ_{exp} and μ_{exp} computed from the first and second halves of the recordings across all datasets. This agreement indicates that slow nonstationarities do not significantly affect the observables considered here.

Partition function of E_{dist} —Here we compute the partition function of the model that matches the mean activity of individual neurons and the distribution along a selected projection, leading to Eq. (7). The partition function is defined as,

$$Z_{\text{dist}} \equiv \sum_s \exp \left[\sum_{n=1}^N h_n s_n - NU \left(\sum_{n=1}^N \frac{W_n}{\sqrt{N}} s_n \right) \right]. \quad (\text{B1})$$

To compute the partition function Z_{dist} we first introduce the following integral,

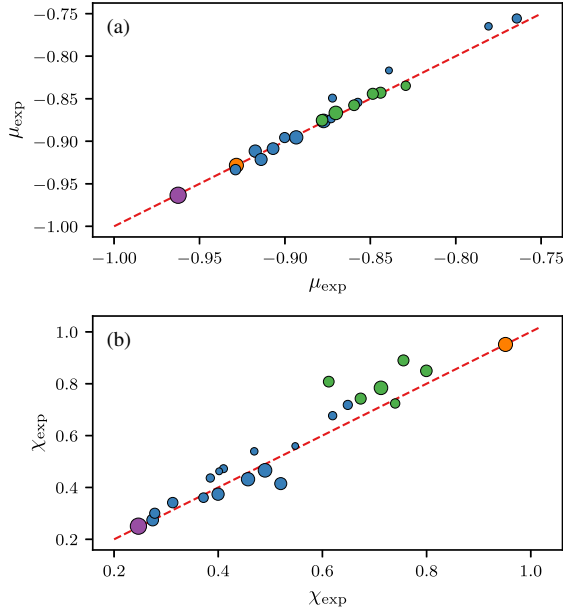


FIG. 6. Consistency of observables across temporal segments. Each dataset was divided into two consecutive, equally sized temporal segments. The experimentally measured observables μ_{exp} (a) and χ_{exp} (b) were computed independently for each segment. Colors and sizes identify datasets and follow the same color code used in Fig. 1 of the main text. The strong correlation along the identity line indicates that these observables are stable across time.

$$Z_{\text{dist}} = \sum_s \int d\varphi \delta\left(\varphi - \sum_n \frac{W_n}{\sqrt{N}} s_n\right) \exp[-E_{\text{dist}}(s)]. \quad (\text{B2})$$

This allows us to rewrite the complicated dependency on the configuration s of $U\left(\sum_n s_n W_n / \sqrt{N}\right)$ as $U(\varphi)$. Then, we use the integral representation of the delta function to decouple the variables $\{s_n\}$, leading to,

$$Z_{\text{dist}} = \sum_s \int \frac{dz}{2\pi} \int d\varphi \exp^{-NU(\varphi) - iz\varphi + \sum_n s_n \left(h_n + iz \frac{W_n}{\sqrt{N}}\right)}. \quad (\text{B3})$$

Summing over the 2^N configurations $\{s_n\}$ gives the integral expression appearing in Eq. (17) of the main text.

Computation of the entropy—The entropy for the distributional model in Eq. (7) is

$$S_{\text{dist}} = \ln Z_{\text{dist}} + \langle E_{\text{dist}}(s) \rangle. \quad (\text{C1})$$

The calculation of the average energy is straightforward

$$\langle E_{\text{dist}} \rangle = \sum_n h_n \mu_n + N \langle U(\varphi) \rangle. \quad (\text{C2})$$

To leading order in $1/N$ the partition function is,

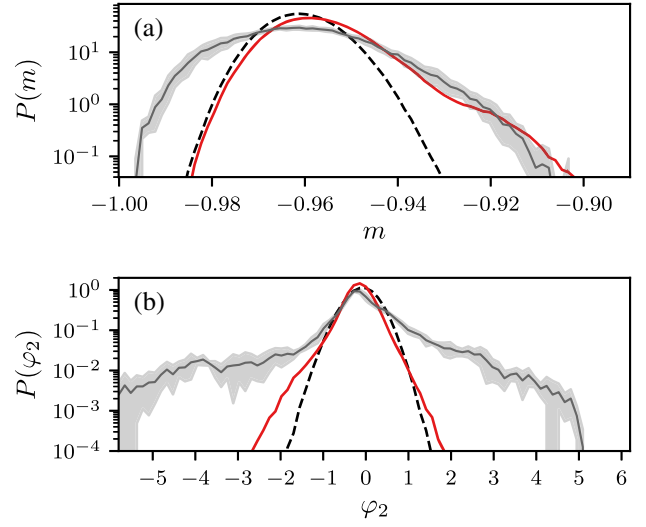


FIG. 7. Testing the maximum entropy model that matches the distribution of neural activity projected along the highest variance eigenvector of the correlation matrix. (a) Distribution of the summed population activity: experimental data (gray), model prediction (red), and the independent model (dashed black). The model accurately captures the broad, non-Gaussian right tail of the distribution. (b) Distribution of neural activity projected on the second principal component. The model (red) provides a slightly better fit than the independent model (dashed black), but fails to capture much of the experimental variance.

$$\ln Z_{\text{dist}} = N \ln 2 - N f_{\text{dist}}(\varphi_{\text{sp}}, z_{\text{sp}}) - \frac{1}{2} \ln \det H + \mathcal{O}(1/N), \quad (\text{C3})$$

where H is the Hessian, or the matrix of second derivatives of $f_{\text{dist}}(\varphi, z)$ evaluated at the saddle point,

$$H = \begin{pmatrix} \frac{1}{N} \sum_n W_n^2 \left[1 - \tanh^2 \left(h_n + i \frac{W_n}{\sqrt{N}} z_{\text{sp}} \right) \right] & i \\ i & N U''(\varphi_{\text{sp}}) \end{pmatrix}. \quad (\text{C4})$$

The curvature of the free energy, given by determinant of H ,

$$\det H = 1 + U'''(\varphi_{\text{sp}}) \sum_n W_n^2 \left[1 - \tanh^2 \left(h_n + i \frac{W_n}{\sqrt{N}} z_{\text{sp}} \right) \right] \quad (\text{C5})$$

rules the stability of the saddle point approximation, with $\det H = 0$ corresponding to a critical transition.

Using the mean-field approximation to leading order in $1/N$ from Eq. (C3), we have

$$S_{\text{dist}} \simeq N \ln 2 - N f_{\text{dist}}(\varphi_{\text{sp}}, z_{\text{sp}}) - \frac{1}{2} \ln \det H - \sum_n h_n \mu_n + N \langle U(\varphi) \rangle. \quad (\text{C6})$$

If we isolate the terms corresponding to the entropy of the independent model, $S_0 = N \ln 2 + \sum_n [\ln \cosh(h_n) - h_n \mu_n]$, the entropy difference $\delta S_{\text{dist}} = S_{\text{dist}} - S_0$ is,

$$\delta S_{\text{dist}} \simeq N [\langle U(\varphi) \rangle - U(\varphi_{\text{sp}})] - \frac{1}{2} \ln [1 + N U''(\varphi_{\text{sp}}) \Delta], \quad (\text{C7})$$

where Δ is defined as in Eq. (27). Finally, the inverse of the $\det H$ is related to the susceptibility χ ,

$$\chi = \frac{\Delta}{1 + N U''(\varphi_{\text{sp}}) \Delta} \quad (\text{C8})$$

Comparison with numerical simulations—Here we test to what extent the model discussed in the main text can capture features beyond the observables constrained in the maximum entropy construction. We consider the distribution of summed population activity $m = (1/N) \sum s_n$, and the distribution of activity along the projection onto the second eigenvector of the covariance matrix; results are in Figs. 7(a) and 7(b), respectively.

We see in Fig. 7(a) that our model does a good job of describing experimental distribution of summed population activity. In particular it accurately reproduces the highly non-Gaussian right-hand tail, corresponding to a huge excess of high activity states relative to what one would expect if neurons were independent. This match extends out to states in which $\sim 4.2\%$ of neurons are active simultaneously, which happens only 0.038% of the time. This success is not just because the weights W overlap the uniform vector, since randomizing the components of W preserves this overlap but spoils the agreement.

In contrast to the case of the summed activity, the model does a very bad job of predicting the distribution of activity along the second principal component [Fig 7(b)]. Indeed,

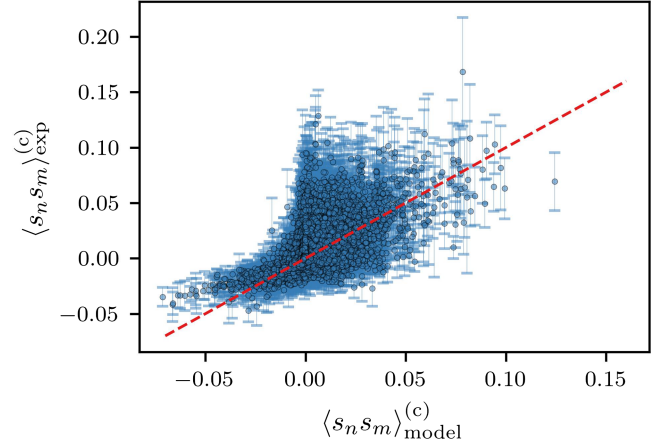


FIG. 8. Connected pairwise correlations. Model vs experimental data, with error bars from variations across fifths of the data. Although pairwise correlations were not explicitly constrained, the model successfully reproduces the overall trend and captures several of the large entries of the correlation matrix.

the predicted distribution is very similar to what we would see if the neurons were completely independent. This is perhaps not surprising, since the principal components are by definition uncorrelated at the leading order, and so we expect that knowing something about one component is relatively uninformative about other components; this is not exactly true because the distributions are not Gaussian.

Despite having used the covariance matrix just to choose the direction with maximum variance, the nonquadratic form of the potential $U(\varphi)$ allows us to make nontrivial predictions for all $\sim N^2/2$ of the correlations. As shown in Fig. 8 the model captures the overall trend of the experimental data and reproduces several of the large entries in the correlation matrix within error bars, despite these not being explicitly constrained. We emphasize that this is not because the covariance matrix is of low rank—the naïve approximation $C \sim \Delta W W^T$ fails completely. We conclude that the nonquadratic terms in $U(\phi)$ are making it possible for a model that focuses on a single projection to make rough but nontrivial predictions beyond the rank one approximation to the covariance.